

Quantum pseudocontexts: Equal rank-one decompositions without hypergraph forcing

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We study a quantum, coordinatization-dependent notion of pseudocontext. Two disjoint families of rays form a quantum pseudocontext when their rank-one projectors have the same sum. Equivalently, the two Born-probability sums agree for every density operator. No finite orthogonality hypergraph is required to force the identity, and probabilities not generated by the given projectors need not respect it. We characterize the construction as a pair of unit-norm rank-one decompositions of one positive structure operator. Complex examples exist for every size $k \geq 2$, whereas real examples first occur at $k = 3$; the size-two separation is caused by a complex relative phase. In dimension two, the Bloch map identifies quantum pseudocontexts with disjoint point sets having equal centroids. In arbitrary dimension, Gram matrices, the Schur–Horn theorem, and the commutant of the structure operator describe the decomposition space. Tight frames generate continuous real families by rotation, including a full-support tetrahedral example in $\mathbb{R}^3$. We distinguish equality of the two quantum statistics, which is state-independent, from constancy of their common value, which occurs only for a scalar structure operator on its support.

I. INTRODUCTION

Let $a_1, \dots, a_k$ and $b_1, \dots, b_k$ be rays in a finite-dimensional real or complex Hilbert space. The identity

$$\sum_{i=1}^k P_{a_i} = \sum_{i=1}^k P_{b_i} \quad (1)$$

between their rank-one projectors implies

$$\sum_{i=1}^k \text{tr}(\rho P_{a_i}) = \sum_{i=1}^k \text{tr}(\rho P_{b_i}) \quad (2)$$

for every density operator ρ . The two collections need not be orthogonal, jointly measurable, or bases. They are two different quantum descriptions of the same positive operator.

Families with this property arose in the study of probabilistic equivalence between noncommuting observables [1, 2]. Two notions should now be separated. In the stronger, combinatorial notion, a finite set of context-normalization equations forces the equality, independently of a faithful orthogonal representation and of the probability theory [3]. The present paper studies the weaker quantum notion in Eq. (1). The coordinates, or equivalently the projectors, are part of the data. The equality is universal over Born-rule states for those data but need not hold after a recoordination or for classical, softmax-generated, or other probability assignments.

This weaker notion has its own natural theory. It is a problem about equal unit-norm rank-one decompositions, hence about finite frames and positive operators. We establish the sharp minimum sizes over $\mathbb{R}$ and $\mathbb{C}$, give continuous constructions for every allowed size, describe the complete two-dimensional geometry, and organize the general case by the spectrum and commutant of the common operator. A four-ray example will also show directly that quantum equality alone does not imply hypergraph or classical equality.

II. DEFINITION AND OPERATOR EQUIVALENCE

Throughout, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and H is a finite-dimensional Hilbert space over $\mathbb{F}$. A nonzero vector a determines a ray $[a] = \mathbb{F}a$ and a projector

$$P_a = \frac{|a\rangle\langle a|}{\langle a, a \rangle}. \quad (3)$$

Thus P_a depends only on the ray. For a density operator ρ , the Born probability of this ray is

$$p_\rho(a) = \text{tr}(\rho P_a), \quad \rho \geq 0, \quad \text{tr} \rho = 1. \quad (4)$$

In dimension at least three these are precisely the finitely additive Hilbert-lattice states under the hypotheses of Gleason’s theorem [4]. In dimension two we deliberately restrict attention to density-operator states.

Lemma 1 (Separation by quantum states). *If self-adjoint operators X, Y satisfy $\text{tr}(\rho X) = \text{tr}(\rho Y)$ for every density operator ρ , then $X = Y$.*

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Proof. Set $C = X - Y$ and take $\rho = P_x$ for every unit vector x . Then $\langle x, Cx \rangle = 0$. The real or complex polarization identity shows that every matrix element of the self-adjoint operator C vanishes. $\square$

Definition 2 (Quantum pseudocontext). A *quantum pseudocontext of size k* in H is a pair of k -element ray families

$$A = \{[a_1], \dots, [a_k]\}, \quad B = \{[b_1], \dots, [b_k]\}$$

such that the rays within each family are distinct, $A \cap B = \emptyset$, and

$$\sum_{i=1}^k p_\rho(a_i) = \sum_{i=1}^k p_\rho(b_i) \quad \text{for every density operator } \rho. \quad (5)$$

It is *genuine* if neither family is an orthonormal basis of its supporting subspace.

Disjointness prevents trivial padding by rays common to both sides. A basis of the common support will be called a relative context; it need not be a basis of the ambient Hilbert space.

By Lemma 1, Definition 2 is equivalent to Eq. (1). We call the common positive operator

$$T := \sum_{i=1}^k P_{a_i} = \sum_{i=1}^k P_{b_i} \quad (6)$$

the *structure operator*. In particular,

$$T \geq 0, \quad \text{tr } T = k, \quad \text{rank } T \leq k. \quad (7)$$

Proposition 3 (Genuineness criterion). *For a quantum pseudocontext with structure operator T , the following are equivalent:*

1. *the pseudocontext is genuine;*
2. *T is not the orthogonal projector onto $\text{ran } T$;*
3. $\lambda_{\max}(T) > 1$.

Proof. One family is an orthonormal basis of $\text{ran } T$ exactly when its projector sum is $P_{\text{ran } T}$. By Eq. (6), this then holds for both families. If all eigenvalues of T were at most one, then

$$k = \text{tr } T \leq \text{rank } T \leq k.$$

Equality throughout forces exactly k nonzero eigenvalues, all equal to one, so $T = P_{\text{ran } T}$. The converse is immediate. $\square$

Consequently, a genuine quantum pseudocontext equates two Born-weight sums that can exceed one. The equality of the sums is state-independent; their common numerical value $\text{tr}(\rho T)$ is generally state-dependent.

III. QUANTUM EQUALITY NEED NOT BE CLASSICALLY FORCED

The smallest complex example already separates the quantum identity from all hypergraph-free probability assignments.

Example 4 (A non-forced four-ray identity). In $\mathbb{C}^2$, let

$$u_\pm = \left(\frac{\sqrt{3}}{2}, \pm \frac{1}{2} \right), \quad v_\pm = \left(\frac{\sqrt{3}}{2}, \pm \frac{i}{2} \right). \quad (8)$$

The four rays are distinct and

$$P_{u_+} + P_{u_-} = P_{v_+} + P_{v_-} = \text{diag} \left(\frac{3}{2}, \frac{1}{2} \right). \quad (9)$$

Both inner products within a target pair equal $1/2$, so the quantum pseudocontext is genuine.

Every pair among the four rays is nonorthogonal: the cross inner products are $3/4 \pm i/4$. Their bare orthogonality hypergraph therefore has no contexts and imposes no normalization equation. The assignment

$$q(u_+) = q(u_-) = 1, \quad q(v_+) = q(v_-) = 0 \quad (10)$$

is a legitimate collection of numbers in $[0, 1]$ on that bare structure, yet its two sums are 2 and 0. A generic softmax or other parametrization can likewise violate the equality. Equation (9) is guaranteed only when the probabilities are produced from the displayed projectors by the Born rule, or by another model that separately imposes the same operator identity.

This example also shows why quantifying over every density operator does not make the identity coordinatization-independent. Moving the four rays while preserving their empty orthogonality graph will generically destroy Eq. (9).

IV. DIMENSION TWO: BLOCH GEOMETRY AND SHARP THRESHOLDS

Every rank-one projector in $\mathbb{C}^2$ has the Bloch form

$$P_a = \frac{1}{2}(I + \mathbf{r}_a \cdot \boldsymbol{\sigma}), \quad \mathbf{r}_a \in S^2 \subset \mathbb{R}^3, \quad (11)$$

where $\boldsymbol{\sigma}$ denotes the Pauli vector.

Theorem 5 (Bloch-sphere characterization). *Two disjoint k -element ray families in $\mathbb{C}^2$ form a quantum pseudocontext if and only if their Bloch vectors have the same sum:*

$$\sum_{i=1}^k \mathbf{r}_{a_i} = \sum_{i=1}^k \mathbf{r}_{b_i}. \quad (12)$$

The structure operator is

$$T = \frac{k}{2}I + \frac{1}{2}\mathbf{R} \cdot \boldsymbol{\sigma}, \quad \mathbf{R} = \sum_i \mathbf{r}_{a_i}, \quad (13)$$

and has eigenvalues

$$\lambda_{\pm} = \frac{k \pm \|\mathbf{R}\|}{2}. \quad (14)$$

Proof. Substitution of Eq. (11) into the two projector sums proves Eq. (12), because $I, \sigma_x, \sigma_y, \sigma_z$ are linearly independent. The Pauli identity $(\mathbf{R} \cdot \boldsymbol{\sigma})^2 = \|\mathbf{R}\|^2 I$ gives the spectrum. $\square$

For $k = 2$ and $0 < \|\mathbf{R}\| < 2$, write

$$\mathbf{r}_{1,2} = \frac{\mathbf{R}}{2} \pm \mathbf{w}, \quad \mathbf{w} \perp \mathbf{R}, \quad \|\mathbf{w}\|^2 = 1 - \frac{\|\mathbf{R}\|^2}{4}. \quad (15)$$

Thus the unordered two-point decompositions of a fixed nonzero $\mathbf{R}$ form a circle. Example 4 selects two points on this circle of decompositions.

The real projective line is different. A real ray represented by $x_{\theta} = (\cos \theta, \sin \theta)$ has

$$P_{\theta} = \frac{1}{2} \begin{pmatrix} 1 + \cos 2\theta & \sin 2\theta \\ \sin 2\theta & 1 - \cos 2\theta \end{pmatrix}. \quad (16)$$

It is therefore represented by a point $q_{\theta} = (\cos 2\theta, \sin 2\theta)$ on a circle rather than by the full Bloch sphere.

Proposition 6 (Real rigidity at size two). *No genuine quantum pseudocontext of size $k = 2$ exists in a real finite-dimensional Hilbert space.*

Proof. The common range of two equal two-projector sums has dimension at most two, so it is enough to work in $\mathbb{R}^2$. Equality is equivalent, by Eq. (16), to equality of two sums of unit vectors on the doubled-angle circle. A nonzero vector has a unique decomposition into two unit circle points, up to permutation. Hence the two ray pairs coincide. If the common sum is zero, each pair is antipodal on the doubled-angle circle, which means that the original rays are orthogonal. Both pairs are then relative contexts and the construction is not genuine. $\square$

The geometric reason is codimension: the complex Bloch sphere offers a circle of chords with a fixed midpoint, while the real Bloch circle offers only one such chord.

V. CONSTRUCTIONS OF EVERY ALLOWED SIZE

The complex size-two phase freedom extends uniformly to all sizes.

Theorem 7 (Complex phase construction). *Let H contain an orthonormal pair e_1, e_2 , let $k \geq 2$, and put $\omega = e^{2\pi i/k}$. Choose $\alpha \in (0, \pi/2)$ and a phase $e^{i\phi}$ that is not a k th root of unity. For $j = 0, \dots, k-1$, define*

$$\begin{aligned} a_j &= \cos \alpha e_1 + \sin \alpha \omega^j e_2, \\ b_j &= \cos \alpha e_1 + \sin \alpha e^{i\phi} \omega^j e_2. \end{aligned} \quad (17)$$

Then all $2k$ rays are distinct and

$$\sum_{j=0}^{k-1} P_{a_j} = \sum_{j=0}^{k-1} P_{b_j} = k(\cos^2 \alpha P_{e_1} + \sin^2 \alpha P_{e_2}). \quad (18)$$

The quantum pseudocontext is genuine whenever $k \geq 3$, and also for $k = 2$ provided $\alpha \neq \pi/4$.

Proof. In each projector, the diagonal terms are $\cos^2 \alpha P_{e_1} + \sin^2 \alpha P_{e_2}$. The off-diagonal terms are proportional to ω^j or ω^{-j} and vanish upon summation because $\sum_{j=0}^{k-1} \omega^j = 0$. The additional phase in the b -family does not affect this cancellation.

Comparison of the nonzero e_1 components shows that two vectors in Eq. (17) can span the same ray only if their second components agree. Powers of ω distinguish rays within each family, and the assumption on $e^{i\phi}$ excludes cross-family collisions. For $k \geq 3$, a k -element family in a two-dimensional support cannot be a basis. For $k = 2$, the inner product within each pair is $\cos 2\alpha$, which is nonzero exactly when $\alpha \neq \pi/4$. $\square$

At $\alpha = \pi/4$, the common operator is $(k/2)I$ on the support. For $k \geq 3$ this is a pair of unit-norm tight frames; for $k = 2$ both families are orthonormal bases and hence nongenuine.

Real examples exist from size three onward. Let

$$\begin{aligned} \theta_j &= \theta_0 + \frac{\pi j}{k}, \quad x_j = (\cos \theta_j, \sin \theta_j), \\ j &= 0, \dots, k-1. \end{aligned} \quad (19)$$

The doubled-angle points form a regular k -gon and sum to zero. A second copy rotated by $\delta \not\equiv m\pi/k \pmod{\pi}$ is ray-disjoint and has the same zero centroid. Therefore both projector sums equal $(k/2)I_2$. For $k \geq 3$ the two families form a genuine real quantum pseudocontext.

Example 8 (Mercedes-Benz pair). For $k = 3$, take ray directions $0, 2\pi/3, 4\pi/3$ and rotate the second triple by an angle θ that produces no ray collision. Both sums equal $(3/2)I_2$.

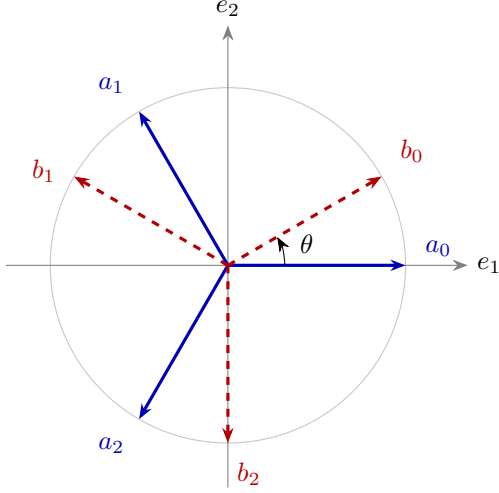


FIG. 1. A real size-three quantum pseudocontext, drawn for $\theta = \pi/6$. The two triples are distinct tight frames with common structure operator $(3/2)I_2$.

Combining these constructions with Proposition 6 gives the sharp field-dependent threshold.

Corollary 9 (Minimum genuine size). *The minimum size of a genuine quantum pseudocontext is 2 over $\mathbb{C}$ and 3 over $\mathbb{R}$.*

No size-one example exists over either field, because equality of two rank-one projectors means equality of their rays, contradicting disjointness.

VI. STRUCTURE OPERATORS, GRAM MATRICES, AND SCHUR–HORN THEORY

Let $A : \mathbb{F}^k \rightarrow H$ be the synthesis operator whose columns are unit vectors $a_1, \dots, a_k$. Its frame operator and Gram matrix are

$$T = AA^\dagger, \quad G = A^\dagger A = (\langle a_i, a_j \rangle)_{i,j=1}^k. \quad (20)$$

The nonzero spectra of T and G agree, including multiplicities. In particular,

$$\text{tr } T = k, \quad (21)$$

$$\text{tr}(T^2) = \sum_{i,j=1}^k |\langle a_i, a_j \rangle|^2 = k + \sum_{i \neq j} |\langle a_i, a_j \rangle|^2. \quad (22)$$

Equation (22) gives another proof that a nonorthogonal family has $\lambda_{\max}(T) > 1$.

Which positive operators occur as sums of k unit rank-one projectors? The answer is especially simple.

Theorem 10 (Unit-norm rank-one decompositions). *A positive operator T admits a decomposition*

$$T = \sum_{j=1}^k P_{a_j}, \quad \|a_j\| = 1, \quad (23)$$

if and only if

$$\text{tr } T = k, \quad \text{rank } T \leq k. \quad (24)$$

This statement allows repeated rays; distinctness is an additional condition for quantum pseudocontexts.

Proof. Necessity is Eq. (7). Conversely, let the positive eigenvalues of T be $\lambda_1 \geq \dots \geq \lambda_r > 0$ and pad this list with $k - r$ zeros. The resulting decreasing k -vector has average one, hence majorizes $(1, \dots, 1)$. By the Schur–Horn theorem [5], there is a positive semidefinite $k \times k$ Gram matrix with this spectrum and diagonal $(1, \dots, 1)$. Factoring the Gram matrix produces k unit vectors whose frame operator has the prescribed nonzero spectrum; an isometry identifies it with T . $\square$

Fix an eigenbasis of T on its support and write $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_r)$. Every decomposition corresponds to a matrix $M \in \mathbb{F}^{r \times k}$ satisfying

$$MM^\dagger = \Lambda, \quad \|m_j\| = 1 \quad (j = 1, \dots, k), \quad (25)$$

where m_j is its j th column. Equivalently,

$$M = \Lambda^{1/2}V, \quad VV^\dagger = I_r, \quad \text{diag}(V^\dagger \Lambda V) = \mathbf{1}. \quad (26)$$

A quantum pseudocontext is a pair of points in this decomposition space whose ray sets are internally distinct and mutually disjoint, modulo column phases (signs over $\mathbb{R}$) and permutations.

The eigenvalues of T also give the exact range of the common statistic:

$$\lambda_{\min}(T) \leq \text{tr}(\rho T) \leq \lambda_{\max}(T), \quad (27)$$

where the endpoints are attained on eigenstates. If the ambient space is larger than $\text{supp } T$, then $\lambda_{\min}(T) = 0$ on the full space. On the support, the lower endpoint is the smallest positive eigenvalue.

VII. SYMMETRY AND FULL-SUPPORT REAL FAMILIES

If U is unitary (orthogonal over $\mathbb{R}$) and commutes with T , then

$$\sum_i P_{Ua_i} = U \left(\sum_i P_{a_i} \right) U^\dagger = T. \quad (28)$$

Thus the isometric commutant

$$\mathcal{C}_{\mathbb{F}}(T) = \{U : U^\dagger U = I, UT = TU\} \quad (29)$$

acts on the decomposition space. If the distinct eigenspaces of T have dimensions m_α , then

$$\mathcal{C}_{\mathbb{C}}(T) \cong \prod_{\alpha} U(m_\alpha), \quad \mathcal{C}_{\mathbb{R}}(T) \cong \prod_{\alpha} O(m_\alpha). \quad (30)$$

Positive spectral degeneracy therefore supplies continuous spatial freedom. Complex phases leave continuous freedom even for simple positive spectra, whereas the corresponding real commutant on the support is discrete.

The maximally symmetric case is a unit-norm tight frame:

$$T = \sum_{i=1}^k P_{a_i} = \frac{k}{n} I_n. \quad (31)$$

Every $R \in O(n)$ preserves this operator. Apart from the finite union of subvarieties on which some Ra_j collides with an original ray, a rotation therefore gives a disjoint second decomposition. If $k > n$, neither family can be a basis, so the resulting quantum pseudocontext is genuine.

Example 11 (Tetrahedral quantum pseudocontext). In $\mathbb{R}^3$, let

$$\begin{aligned} a_1 &= (0, 0, 1), \\ a_2 &= \left(\frac{2\sqrt{2}}{3}, 0, -\frac{1}{3} \right), \\ a_3 &= \left(-\frac{\sqrt{2}}{3}, \frac{\sqrt{6}}{3}, -\frac{1}{3} \right), \\ a_4 &= \left(-\frac{\sqrt{2}}{3}, -\frac{\sqrt{6}}{3}, -\frac{1}{3} \right). \end{aligned} \quad (32)$$

These vectors form a regular tetrahedral tight frame with

$$\sum_{j=1}^4 P_{a_j} = \frac{4}{3} I_3. \quad (33)$$

Let R be the quarter-turn about the x axis,

$$R = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad b_j = Ra_j. \quad (34)$$

No b_j is collinear with an a_i , and $\sum_j P_{b_j} = (4/3)I_3$. The eight rays therefore form a genuine, full-support real quantum pseudocontext of size four.

The same mechanism applies to any unit-norm tight frame after a collision-free rotation. It does not produce a genuine size-three example spanning $\mathbb{R}^3$: if three unit vectors satisfy $\sum_{i=1}^3 P_{a_i} = I_3$, then Eq. (22) forces all off-diagonal inner products to vanish. They are an orthonormal basis. Full-support real size-three examples must therefore have nonscalar structure operators; concrete instances occur in Ref. [1].

VIII. OPERATIONAL MEANING

The projectors within one family generally do not commute and cannot be measured together as one sharp test.

The statistic

$$\rho \mapsto \sum_i \text{tr}(\rho P_{a_i}) = \text{tr}(\rho T) \quad (35)$$

is estimated from separate compatible measurement arrangements on repeated preparations. Its equivalence with the b -statistic is exact for every quantum state.

There is also a direct positive operator-valued measure (POVM) formulation. Since $0 \leq T \leq kI$, the effects

$$\begin{aligned} E_i &= \frac{1}{k} P_{a_i}, & E_0 &= I - \frac{1}{k} T, \\ F_i &= \frac{1}{k} P_{b_i}, & F_0 &= I - \frac{1}{k} T \end{aligned} \quad (36)$$

define two POVMs. Their non-residual outcomes have the same coarse-grained effect T/k . If $T = (k/r)I$ on an r -dimensional support, the more natural normalization $(r/k)P_{a_i}$ removes the residual effect on that support and gives two rank-one tight-frame POVMs.

This equivalence is related to, but is not by itself, generalized contextuality in the sense of Spekkens [6]. Nor is it a Kochen–Specker contradiction. The four-ray assignment (10) demonstrates that an ontological or classical model is under no obligation to respect the quantum identity unless the equality is added as an operational equivalence or follows independently from its own admissibility constraints.

IX. RELATION TO HYPERGRAPH-FORCED PSEUDOCONTEXTS

In standard graph-theoretic terminology, a *faithful orthogonal representation* (FOR), also called a faithful coordinatization, assigns distinct rays to the vertices so that two rays are orthogonal exactly when the corresponding vertices share a hyperedge [7–9]. Every hypergraph-forced pseudocontext whose hypergraph admits a FOR yields a quantum pseudocontext, because a balanced linear combination of context equations also combines the corresponding basis projector equations. The converse fails. Example 4 has no bare context equations, and a generic tight frame paired with a generic rotation also carries too little bare orthogonality data to imply its projector-sum identity.

Whether a given quantum pseudocontext can be embedded into a larger finite context hypergraph that forces the same equality is an additional completion problem, not part of Definition 2. A complex three-dimensional size-two completion and its field-dependent realizability are studied in Ref. [10]; the general finite-completion question remains open. The logical distinction is:

finite hypergraph forcing

$\Downarrow$

quantum projector equality,

(37)

with no converse in general.

X. OPEN PROBLEMS

Open Problem 12 (Decomposition moduli). For a fixed positive T , determine the connected components and singular strata of the unit-norm decomposition space in Eq. (25), modulo column phases, permutations, and the isometric commutant of T .

Open Problem 13 (Disjoint decomposition criterion). Strengthen Theorem 10 from the existence of one unit-norm decomposition to a spectral or geometric criterion for two ray-disjoint decompositions, over both $\mathbb{R}$ and $\mathbb{C}$.

Open Problem 14 (Finite forcing completion). Characterize those quantum pseudocontexts that admit a finite auxiliary context hypergraph whose normalization equations force their equality. In particular, determine which generic tight-frame rotations have such a completion.

XI. CONCLUSION

Quantum pseudocontexts are pairs of disjoint unit-norm rank-one decompositions of one positive operator. Their defining equality holds for all density operators because quantum states separate self-adjoint operators. This universality is internal to the chosen quantum coordinatization: it does not make the equality a consequence of a bare orthogonality hypergraph and does not constrain arbitrary classical or softmax probabilities.

The distinction exposes a clean mathematical structure. In $\mathbb{C}^2$, equal projector sums are equal Bloch centroids; complex phase freedom permits size two, while real doubled-angle geometry excludes it. Schur–Horn theory describes the possible structure operators, their canonical fibers describe all unit-norm decompositions, and spectral degeneracy generates continuous families through the commutant. These operator-level identities are useful as quantum equivalences in their own right, independently of whether any finite hypergraph can force them.

DATA AVAILABILITY

All coordinates used in the examples are displayed in the text. Lean 4 sources verifying the finite coordinate-operator algebra of the complex phase construction, ray disjointness, and the trace identities are available from the corresponding authors upon reasonable request.

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